

Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital IO module is sturdy, low power usage and easy to use.

8 Port DI and 8 Port DO Module:-



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. The DIP switch for Slave ID and Baud rate are placed inside the enclosure.

The design of the modules incorporates '**resettable Fuses**' to safeguard against sudden High Voltage spikes. The board also has '**reverse polarity**' protection both for input **Power** connector

The communication port also has signal isolation and additional Jumper for 'end of line' 120 ohms termination resistance which may or may not be used.

Specifications

General:-

I/O Connectors	12 Pin 5.08 mm and 2 Pin 5.08 mm pitch pluggable ew terminal block
Dimensions	110 mm L × 110 mm B × 50 mm H
Power	Input Power: 9–36 V DC or 24 V AC/DC Typical: 12 V DC 1 A
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



NOTE:-

Without Load : ~69 mA @ 12 VDC

With Full Load: ~300 mA @ 12 VDC(8 relays ON)

Relay Output:-

Channels	8
Contact Form	1A, 1C
Contact Material	Ag Alloy
Contact Capacity	10A @ 240VAC, 10A @ 28VDC
Coil Voltage	5 – 12 VDC
Coil Power	0.36 W
Insulation Resistance	250MΩ
Electrical Life	1 x 10 ⁵
Mechanical Life	1 x 10 ⁷
Operating Time	7 msec, Max 15 msec
Release Time	2 msec, Max 6 msec

DI Inputs:-

Channels	8
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

Additional Features:-

All inputs and communication port isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air
EFT IEC 61000-4-4, 50A (5/50ms)
400V isolation.
CRC Error check.
No configuration needed on the IO board

DIP SETTING – For Manual Switch Setup

Configuration Settings:-

Communication Speed	9600 - 115200(DIP SW selectable)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	1 – 15 (DIP SW selectable)
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	0,1,2,3,4,5,6,7.
Function code DI	0x02 Read Discrete Input
DI Register Address	8,9,10,11,12,13,14,15.

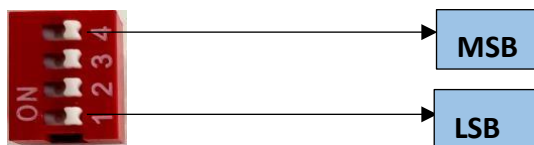
Recommended Cable Electrical Characteristics:-

22 AWG Cable	Shielded and twisted pair
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5 db / 100 ft

Note:-

For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

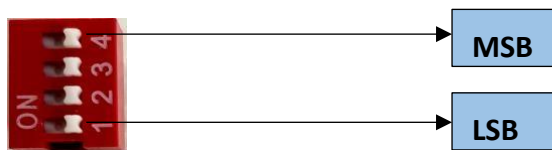
BAUD RATE DESCRIPTION:-



- For Baud rate Selection, DIP SW is used as per the diagram.
- Pulling up the switch will make Baud rate active.
- If no selection is made 9600 will be default Baud rate.
- When you change the Baud rate in the Module power 'ON' condition, please press the reset button to get Change to affect.

Baud Rate	DIP SWITCH			
	1	2	3	4
9600	OFF	OFF	OFF	OFF
19200	OFF	OFF	OFF	ON
38400	OFF	OFF	ON	OFF
57600	OFF	ON	OFF	OFF
115200	ON	OFF	OFF	OFF

SLAVE ID DESCRIPTION:-



- For Slave ID Selection SW is used to Set The SLAVE ID .
- For Slave ID DIP Switch **LSB is “1”** follow through **“4” is MSB**.
- Slave ID Confirmed through below Device ID table .
- IF Eg. Slave ID 1 is Needed to be selected Switch number 1 should pulled up other three should be selected down side. So”1 0 0 0” will be selected as Slave ID 1.

Slave ID	DIP SWITCH				OUTPUT (Binary)	OUTPUT (Decimal)
	1	2	3	4		
0	OFF(0)	OFF(0)	OFF(0)	OFF(0)	1 0 0 0	1
1	ON(1)	OFF(0)	OFF(0)	OFF(0)	1 0 0 0	1
2	OFF(0)	ON(1)	OFF(0)	OFF(0)	0 1 0 0	2
3	ON(1)	ON(1)	OFF(0)	OFF(0)	1 1 0 0	3
4	OFF(0)	OFF(0)	ON(1)	OFF(0)	0 0 1 0	4
5	ON(1)	OFF(0)	ON(1)	OFF(0)	1 0 1 0	5
6	OFF(0)	ON(1)	ON(1)	OFF(0)	0 1 1 0	6
7	ON(1)	ON(1)	ON(1)	OFF(0)	1 1 1 0	7
8	OFF(0)	OFF(0)	OFF(0)	ON(1)	1 0 0 0	8
9	ON(1)	OFF(0)	OFF(0)	ON(1)	1 0 0 1	9
10	OFF(0)	ON(1)	OFF(0)	ON(1)	1 0 1 0	10
11	ON(1)	OFF(0)	ON(1)	ON(1)	1 0 1 1	11
12	OFF(0)	OFF(0)	ON(1)	ON(1)	1 1 0 0	12
13	ON(1)	OFF(0)	ON(1)	ON(1)	1 1 0 1	13
14	OFF(0)	ON(1)	ON(1)	ON(1)	1 1 1 0	14
15	ON(1)	ON(1)	ON(1)	ON(1)	1 1 1 1	15

SOFT SETTING – For Modbus / Software Setup

Configuration Settings:-

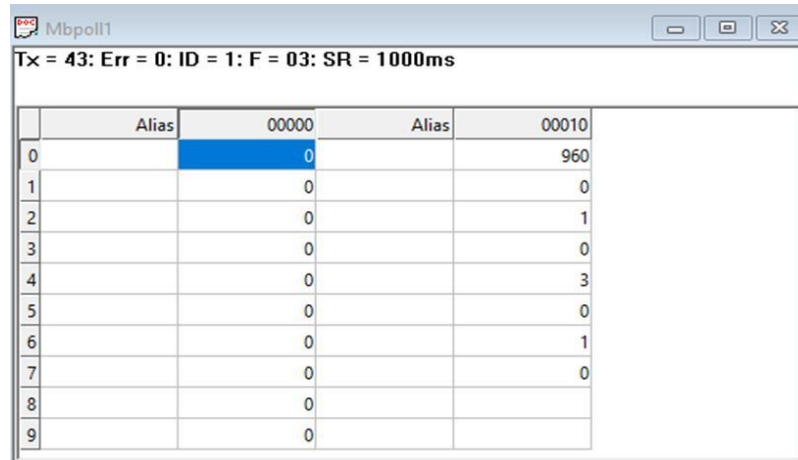
Communication Speed	9600 – 115200.
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Soft Setting(0-247).
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	0,1,2,3,4,5,6,7.
Function code DI	0x02 Read Discrete Input
Function code Soft Serial	0x03 Read Holdings Register
DI Register Address	8,9,10,11,12,13,14,15.
Soft Serial Register Address	11, 12,13,14,15,16,17,18.

Note:-

1. In register 40015- you will able to see the current parity and you can change it in register 40016
2. In register 40017- you will able to see the current stop bit and you can change it in register 40018
3. If you want to enter the baud rate 9600/19200/38400/57600/115200etc.Enter it as 960/1920/5760/11520etc.
4. After that press the reset button or reboot the system.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
0	DO 1	00001	0X05,0X0F	RS485	1 Bit Boolean
1	DO 2	00002	0X05,0X0F	RS485	1 Bit Boolean
2	DO 3	00003	0X05,0X0F	RS485	1 Bit Boolean
3	DO 4	00004	0X05,0X0F	RS485	1 Bit Boolean
4	DO 5	00005	0X05,0X0F	RS485	1 Bit Boolean
5	DO 6	00006	0X05,0X0F	RS485	1 Bit Boolean
6	DO 7	00007	0X05,0X0F	RS485	1 Bit Boolean
7	DO 8	00008	0X05,0X0F	RS485	1 Bit Boolean
8	DI 1(With & Without Potential)	00009	0X02	RS485	1 Bit Boolean
9	DI 2(With & Without Potential)	00010	0X02	RS485	1 Bit Boolean
10	DI 3(With & Without Potential)	00011	0X02	RS485	1 Bit Boolean
11	DI 4(With & Without Potential)	00012	0X02	RS485	1 Bit Boolean
12	DI 5(With & Without Potential)	00013	0X02	RS485	1 Bit Boolean
13	DI 6(With & Without Potential)	00014	0X02	RS485	1 Bit Boolean
14	DI 7(With & Without Potential)	00015	0X02	RS485	1 Bit Boolean
15	DI 8(With & Without Potential)	00016	0X02	RS485	1 Bit Boolean
10	Display Baud Rate (Default:960)	40011	0X03	RS485	16 Bit Unsigned int
11	Enter New Baud Rate	40012	0X03	RS485	16 Bit Unsigned int
12	Display Slave ID (Default:1)	40013	0X03	RS485	16 Bit Unsigned int
13	Enter New Slave ID	40014	0X03	RS485	16 Bit Unsigned int
14	Display Parity (Default: None-3 And Odd-1,Even-2)	40015	0X03	RS485	16 Bit Unsigned int
15	Enter New Parity	40016	0X03	RS485	16 Bit Unsigned int
16	Display Stop Bit (start-2/stop bit-1)	40017	0X03	RS485	16 Bit Unsigned int
17	Enter New Start/ Stop Bit.	40018	0X03	RS485	16 Bit Unsigned int

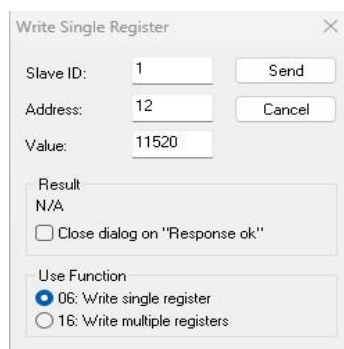
BAUD RATE and SLAVE ID DESCRIPTION:-



Tx = 43: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		960
1		0		0
2		0		1
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

1. As you can see in the image above, the resistor 40011 – 960 represents the present baud rate (9600) of the board and 40013 – 1 represents the present slave id of the board.
2. To change the baud rate and slave id you have to enter a value on the 40012 resistor for the baud rate and 40014 for the slave id as shown in the image below.



Write Single Register

Slave ID: 1 Send

Address: 12 Cancel

Value: 11520

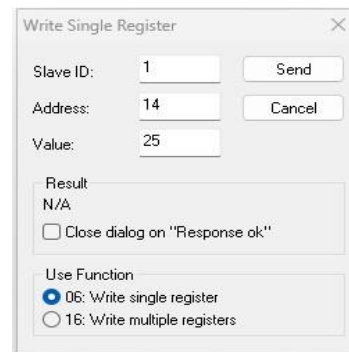
Result: N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register

☐ 16: Write multiple registers



Write Single Register

Slave ID: 1 Send

Address: 14 Cancel

Value: 25

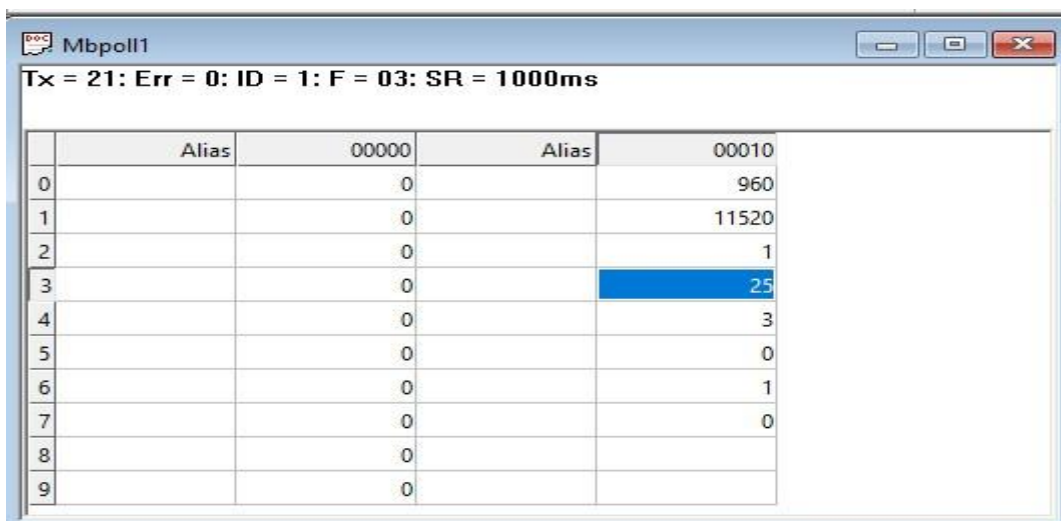
Result: N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register

☐ 16: Write multiple registers



Tx = 21: Err = 0: ID = 1: F = 03: SR = 1000ms

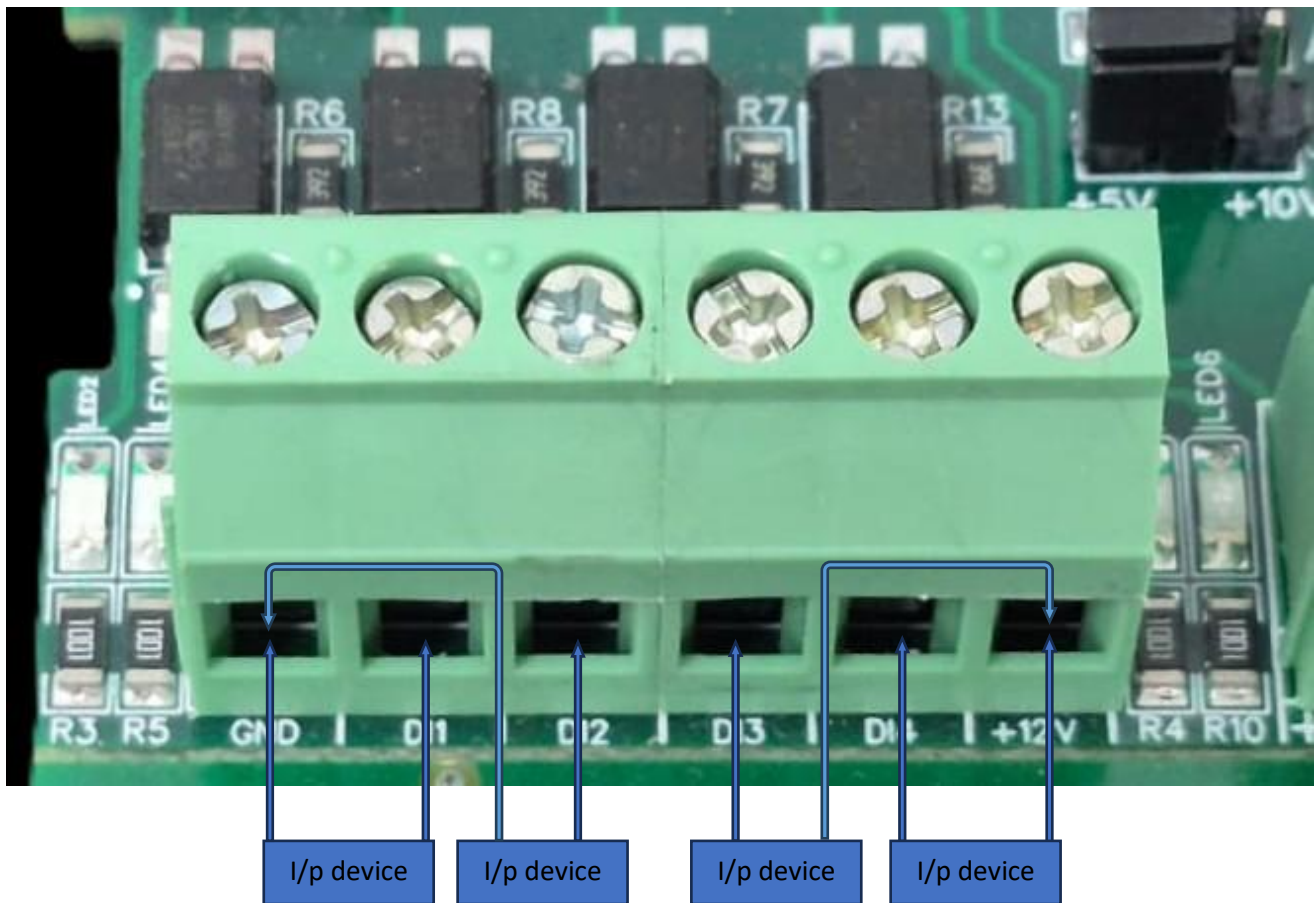
	Alias	00000	Alias	00010
0		0		960
1		0		11520
2		0		1
3		0		25
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above I set 115200 baud rate and slave ID 25.

Note:-

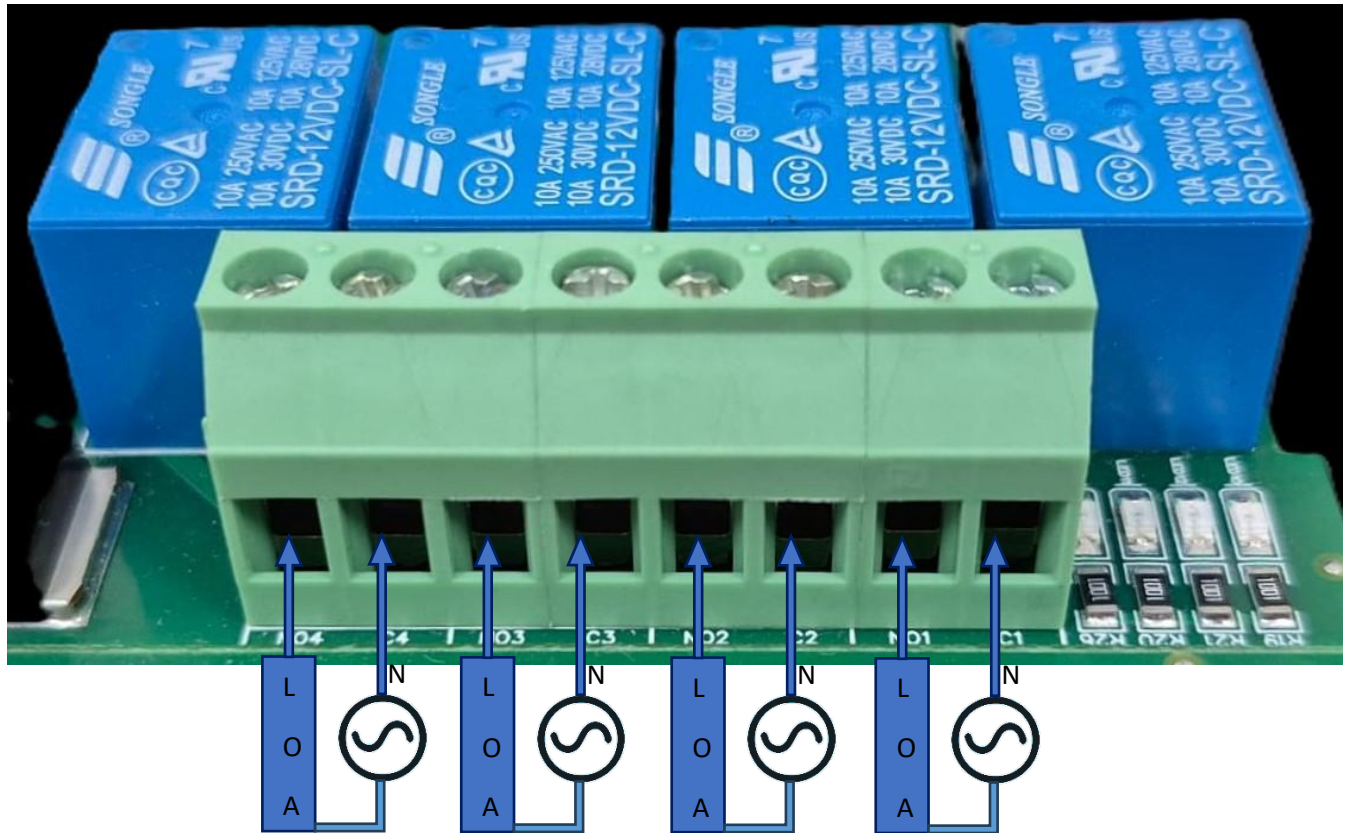
- When entering baud rate values such as **9600 / 19200 / 38400 / 57600 / 115200**, enter them as **960 / 1920 / 3840 / 5760 / 11520**, respectively.
- After entering the values, **press the reset button or reboot the system.**
- Once restarted, you will see the updated values:
- **40011 – 11520 → Baud Rate (115200)**
- **40013 – 25 → Slave ID (25)**
- You can also **change Parity Bit and Stop Bit** of the device in the same way.
- **Jumper Setting (Hardware Method):**
- If you place the **jumper**, the device will **load its default communication settings** (default baud rate(9600) and slave ID(1)).
- This is useful when you have many devices with different settings and want a **quick way to restore defaults** without software changes.
- **Relation between Jumper and Reset Button:**
The **jumper enables the default mode**, and the **reset button activates it**.
- Simply **insert the jumper → press reset → the device restarts with factory default settings** automatically.

DI



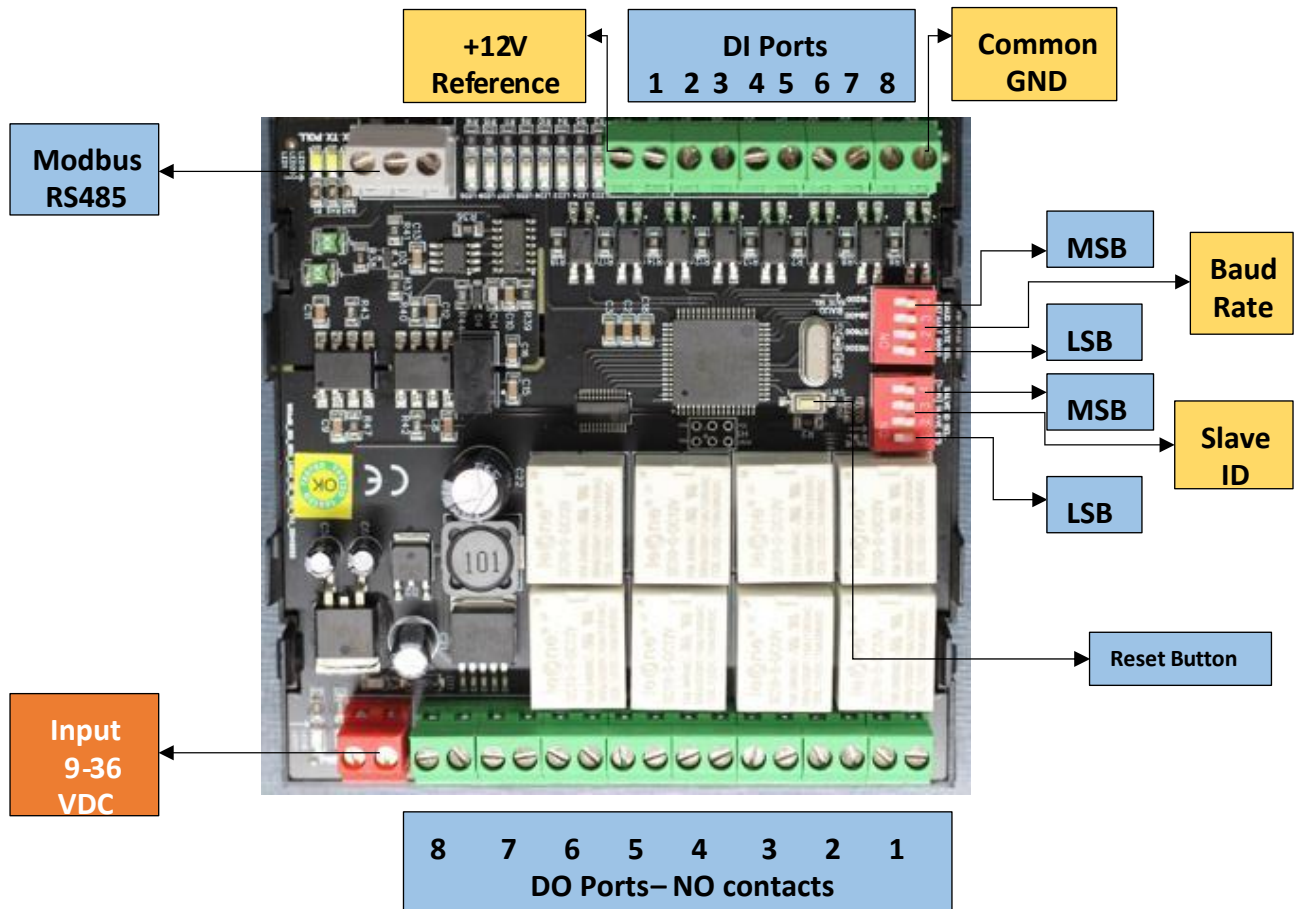
- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is internally connected to **NO**, allowing current to flow. When the relay is de-energized, the connection between **C** and **NO** remains open, and the load stays OFF.
- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The load is connected between terminal **NO** and **Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The load is connected between terminal **NO** and **Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or 10A @ 125VAC) or **10A @ 30VDC** (or 10A @ 28VDC).



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